

INFO8010: Interior Design Style Transfer

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Project: <https://github.com/FredrichT/decogen.git> (main branch)

I. INTRODUCTION

Interior design visualization remains challenging for non-professionals. Most people struggle to imagine how their spaces would look in different design styles. Traditional visualization methods require either professional design skills or extensive photo editing expertise.

In this paper, we explore a machine learning approach for transforming room interior images between design styles. Our method investigates a CycleGAN architecture for style transfer between different interior design aesthetics.

Our work examines:

- Adapting CycleGAN for room style transfer
- Implementing multi-scale discriminators and enhanced residual blocks
- Using asymmetric learning rate scheduling to improve training stability
- Evaluating results using LPIPS and FID metrics

We focus specifically on transferring between coastal and eclectic living room styles. Our approach works with unpaired datasets, addressing the practical limitation that paired before/after images of the same room in different styles are rarely available.

II. RELATED WORK

A. Image-to-Image Translation

Image-to-image translation has seen significant progress in recent years. Pix2Pix [1] introduced a general framework for supervised image translation using conditional GANs, requiring paired training data. CycleGAN [2] extended this approach to the unpaired setting by introducing cycle consistency loss, enabling training without directly corresponding before/after images. This unpaired approach is particularly valuable for interior design applications, where obtaining before/after photos of the same room in different styles is difficult.

B. Style Transfer

Neural style transfer, explored by Gatys et al. [3], showed the ability to transfer artistic styles to photographs using feature representations from pre-trained networks. Subsequent approaches improved efficiency [4] and versatility [5]. In particular, adaptive instance normalization (AdaIN) [5] and whitening and coloring transforms (WCT) [6] demonstrated better separation of content and style. While these methods work well for artistic style transfer, they often struggle with the structural preservation needs of interior design applications.

C. Diffusion Models

More recently, diffusion-based generative models have shown exceptional capabilities in image generation and manipulation tasks. Denoising diffusion probabilistic models (DDPM) [7] introduced a new paradigm for high-quality image synthesis. These techniques have evolved into powerful text-to-image models such as DALL-E 2 [8] and latent diffusion models [9].

While these models can produce impressive style transformations with text guidance, they typically require large-scale training data and computational resources. Some recent work has explored using diffusion models specifically for image-to-image translation tasks [10], though often with different training requirements compared to GAN-based approaches.

D. Interior Design Applications

Image translation and style transfer techniques have been applied to interior design in both academic research and commercial applications. Several existing tools offer interior visualization capabilities.

We chose to work with the CycleGAN architecture for its established ability to work with unpaired image datasets and its accessible computational requirements compared to newer diffusion-based approaches. This makes it well-suited for interior design applications where paired before/after images are rarely available.

III. METHODOLOGY

A. Dataset

Our project uses a curated dataset of living room interior images from the MMIS (Multi-Modal Interior Style) dataset [11]. This dataset provides a comprehensive collection of interior design images with clear style categorizations, specifically focusing on living room spaces.

The dataset characteristics include:

- **Total images:** 2,069 living room images
- **Coastal style:** 1,049 images featuring light, airy spaces with natural materials, white and blue color palettes, and beach-inspired elements
- **Eclectic style:** 1,020 images showcasing bold, mixed patterns, vibrant colors, and diverse decorative elements from various design periods
- **Image quality:** Professional interior photography with consistent lighting and composition
- **Organization:** Images directly organized into style-specific directories (`dataset/coastal/` and `dataset/eclectic/`)

We chose **coastal** and **eclectic** styles due to their distinct visual characteristics. Coastal living rooms typically feature light wood tones, white and blue color palettes, natural textures, and minimalist furniture arrangements. In contrast, eclectic living rooms combine bold colors, mixed patterns, diverse furniture styles, and rich textures. This strong stylistic contrast ensures that style transfer effects are pronounced and easily evaluable.

For training our model, we split each style category into train (70%), validation (15%), and test (15%) sets. This resulted in approximately 734 coastal and 714 eclectic living room images for training, with proportional numbers for validation and testing.

B. Model Architecture

Our architecture is based on the CycleGAN [2] framework. The overall system consists of two generator networks and two discriminator networks, enabling bidirectional translation between coastal and eclectic living room styles. Figure 1 illustrates the high-level architecture of our system.

1. Generator Architecture

Our generator network uses a ResNet-based architecture for living room style transfer:

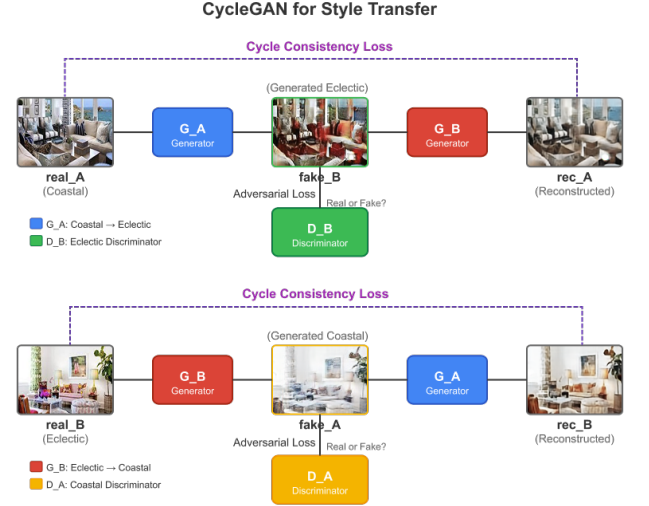


FIG. 1: CycleGAN architecture for living room style transfer between Coastal and Eclectic styles, showing generators (G_A , G_B), discriminators (D_A , D_B), and cycle consistency loss.

- Initial reflection padding and 7×7 convolution to preserve edge information
- Two downsampling layers with stride-2 convolutions and instance normalization
- Nine residual blocks for feature transformation while maintaining spatial information
- Bilinear interpolation upsampling followed by 3×3 convolutions to reduce checkerboard artifacts
- Final reflection padding and 7×7 convolution with Tanh activation

The dual generator structure allows us to create both Coastal $\rightarrow$ Eclectic (G_A) and Eclectic $\rightarrow$ Coastal (G_B) transformations. This bidirectionality enables cycle consistency training without requiring paired images.

2. Multi-scale Discriminator

We implemented a PatchGAN discriminator architecture that operates on image patches rather than the entire image:

- PatchGAN architecture that classifies overlapping patches as real or fake
- Spectral [12] normalization for training stability
- Focuses on local texture and style features rather than global image structure

This approach helps the model distinguish between coastal and eclectic styles by focusing on local style characteristics like textures, patterns, and color combinations typical of each design style.

3. Perceptual Network

C. Loss Functions

Our training objective combines two main loss terms following the CycleGAN approach:

$$\mathcal{L} = \mathcal{L}_{GAN} + \lambda_{cycle}\mathcal{L}_{cycle} \quad (1)$$

1. Adversarial Loss

We use the GAN loss with mean squared error (MSE) for training stability:

$$\mathcal{L}_{GAN}(G, D) = \mathbb{E}_y[(D(y) - 1)^2] + \mathbb{E}_x[D(G(x))^2] \quad (2)$$

2. Cycle Consistency Loss

This loss ensures that transformed images can be mapped back to their original versions:

$$\mathcal{L}_{cycle} = \mathbb{E}_x[||G_B(G_A(x)) - x||_1] + \mathbb{E}_y[||G_A(G_B(y)) - y||_1] \quad (3)$$

This CycleGAN formulation allows the generators to perform substantial style transformations while maintaining cycle consistency, which is essential for unpaired training scenarios where direct supervision is unavailable.

IV. IMPLEMENTATION

A. System Overview

We implemented our system in Python using PyTorch. The system handles the complete pipeline from dataset preparation to inference.

B. Project Structure

Our codebase is organized into several modular components:

- **config.py**: Configuration file containing all hyperparameters, including batch size, learning rates, model dimensions, and loss weights. This configuration file contains the standard parameters which can then be modified by specifying the desired values in the command line.

- **data_preparation.py**: Handles dataset operations including loading bathroom images, splitting into train/validation/test sets (70%/15%/15%), and creating appropriate data loaders with transformations for resizing, cropping, and normalization.
- **train.py**: Implements the training loop with both generator and discriminator optimization steps, periodical checkpoint saving, and sample visualization. Integrated with Weights & Biases to monitor the progress.
- **evaluate.py**: Computes quantitative metrics including LPIPS for measuring perceptual similarity and FID for assessing generation quality. Creates visualizations for qualitative assessment.
- **inference.py**: Applies the trained model to new bathroom images.
- **models/**: Contains the neural network architecture implementation:
 - **cycle_gan.py**: Implements the core CycleGAN model class, coordinating the interaction between generators, discriminators, and multiple loss functions.
 - **networks.py**: Defines the ResNet generator, multi-scale discriminator, and perceptual network architectures.
 - **losses.py**: Implements the loss functions including GAN and cycle consistency losses.
- **utils.py**: Provides utility functions for image processing, tensor-to-image conversion, and visualization.

C. Training Process

We trained our model using the following hyperparameters:

- **Batch size**: 8, providing stable gradient estimates while accommodating memory constraints.
- **Learning rates**: Asymmetric learning rates with generator at 0.0002 and discriminator at 0.0001 using Adam optimizer ($\beta_1 = 0.5$, $\beta_2 = 0.999$). The lower discriminator learning rate prevents it from becoming too dominant during training.
- **Training epochs**: 500, providing sufficient iterations for the model to learn meaningful style transfer between coastal and eclectic designs.
- **Image size**: 256×256, offering a good balance between preserving room details while keeping reasonable computational requirements.

- **Loss weights:** $\lambda_{cycle} = 5.0$. We use only cycle consistency loss alongside the adversarial loss, following a simplified CycleGAN approach that allows more creative freedom in style transformation.

The training process included several strategies for stability:

- **Asymmetric learning rate scheduling:** We implemented different scheduling strategies for generator and discriminator to address the common problem of discriminator dominance in GAN training. The generator uses CosineAnnealingWarmRestarts ($T_0=50$, $T_{mult}=2$) to provide periodic learning rate boosts that help escape local minima. The discriminator uses CosineAnnealingLR with steady decay to prevent it from becoming too powerful relative to the generator.
- **Image buffer of size 50 for discriminator training** to reduce oscillation by using a history of generated images
- **Spectral normalization in the discriminator** to enforce Lipschitz continuity and improve training dynamics
- **Reflection padding to reduce border artifacts**

Weights & Biases allowed us to observe loss curves for adversarial and cycle consistency components, learning rate schedules, and generated samples throughout training.

D. Training Monitoring

We integrated Weights & Biases (wandb) into our training pipeline to track model performance throughout the training. By monitoring various loss components individually, we gained valuable insights into the model’s learning dynamics and convergence patterns.

1. Overall Loss Progression

The validation losses (Figures 9 and 10) reveal important training dynamics. The validation generator loss shows fluctuations throughout training, but after the final warm restart at epoch 350, it demonstrates a stable decreasing trend from approximately 0.7 to 0.4 by epoch 500, indicating the generator was performing better with improved stability. Conversely, the validation discriminator loss shows a concerning upward trend throughout training, rising from approximately 0.25 to 0.35 by epoch 500. This divergence suggests that after the final restart, the generator began learning more effectively while the discriminator’s validation performance degraded. The high validation losses indicate potential overfitting and suggest that longer

training with proper regularization could yield better generalization results. This degradation is also partly due to our deliberate choice of asymmetric learning rates (lower for the discriminator) to prevent the discriminator from outperforming the generator too much. Alternative techniques such as training the discriminator every two epochs instead of every epoch could help maintain better adversarial balance while preserving generator learning capability.

An important observation is that the high variance observed for the validation losses at the beginning of training progressively decreased throughout the training process. By the end of training, we achieved both lower variance and a decreasing trend for the validation generator loss, making later epoch models more reliable. Given the stable decreasing trend observed in the final epochs, longer training could potentially lead to better results.

2. Learning Rate Schedule

The generator and discriminator learning rates (Figures 11 and 12) show our asymmetric scheduling approach. The generator learning rate follows CosineAnnealingWarmRestarts with clear periodic restarts at epochs 50, 150, and 350, providing learning rate boosts that help maintain generative diversity. The discriminator learning rate follows a steady exponential decay from 0.0001 to near zero, preventing discriminator dominance while maintaining training stability.

3. Domain-Specific GAN and Discriminator Losses

The domain-specific GAN losses (Figures 13 and 14) show distinct behaviors for each domain. Domain A shows an upward trend after epoch 300, peaking at around 0.55 at epoch 350 (the final warm restart), but then decreases back to approximately 0.4 by the end of training, indicating improved generator performance after the restart. Domain B shows more stable performance with gradual improvement until epoch 400, followed by slight deterioration.

The discriminator losses for both domains (Figures 15 and 16) show interesting dynamics. Both domain A and B discriminator losses decrease steadily until the final warm restart at epoch 350, reaching their lowest values around 0.1. However, after the restart, both discriminator losses begin to increase, indicating that the improved generator performance after the restart made it more challenging for the discriminators to distinguish real from fake images.

4. Cycle Consistency

The cycle consistency losses (Figures 17 and 18) demonstrate effective learning of the bidirectional mapping. Both domains show strong convergence, starting from values above 1.0 and decreasing to approximately 0.5-0.6 by training completion. The cycle losses shows periodic spikes corresponding to the generator learning rate restarts, but show consistent recovery and improvement.

E. Evaluation Metrics

We evaluated our model using two complementary metrics:

- **LPIPS** (Learned Perceptual Image Patch Similarity): This perceptual metric measures similarity between images based on features extracted from pre-trained networks. We used LPIPS to evaluate two aspects:
 - Cycle consistency: How well the original image is reconstructed after a full translation cycle ($A \rightarrow B \rightarrow A$)
 - Perceptual similarity: How perceptually similar the style-transferred images are to images from the target domain
- **FID** (Fréchet Inception Distance): This metric measures the distance between feature distributions of real and generated images using the Inception-v3 network. Lower FID scores indicate generated images more similar to real ones, providing an assessment of overall generation quality and style transfer effectiveness.

These quantitative metrics, combined with qualitative visual assessment, allowed us to evaluate the performance of our bathroom style transfer system.

V. EXPERIMENTS AND RESULTS

We conducted an evaluation of our style transfer model, inspecting both quantitative metrics and qualitative visual results. Our evaluation focused on the bidirectional style transfer between coastal and eclectic living room styles. We selected the model from epoch 470 for evaluation as it exhibited one of the lowest validation generator losses after the final warm restart and produced visually superior sample results. While this choice could be debated—the overall lowest validation generator loss occurred at epoch 86 (0.184), suggesting we could have used the closest saved model (epoch 85)—that earlier model showed high validation variance (0.41), reflecting the instability characteristic of early training phases. The high

variance observed at the beginning of training progressively decreased, and by the end of training we achieved both lower variance and a decreasing loss trend, making the epoch 470 model more reliable for evaluation.

A. Quantitative Evaluation

Table I presents the quantitative evaluation metrics of our style transfer model after training for 500 epochs (evaluated at epoch 470).

Metric	Value
LPIPS Cycle Consistency A	0.3768
LPIPS Cycle Consistency B	0.3868
LPIPS Average Cycle Consistency	0.3818
LPIPS Perceptual A to B	0.4250
LPIPS Perceptual B to A	0.3838
LPIPS Average Perceptual	0.4044
FID A to B	127.0767
FID B to A	152.0945
FID Average	139.5856

TABLE I: Evaluation metrics for our living room style transfer model at epoch 470.

The LPIPS scores measure perceptual similarity (lower is better), with our final model achieving cycle consistency scores around 0.38. Our model achieves reasonable cycle consistency with an average score of 0.3818, indicating that images can be transformed to the target style and back while maintaining reasonable content preservation. The perceptual transfer scores (0.4044 average) are slightly higher, which is expected as they measure the perceptual difference between domains—indicating meaningful style transfer is occurring.

The FID scores measure the distribution distance between real and generated images, with lower scores indicating better quality. Our model achieved FID scores of 127.08 for Coastal to Eclectic transfer and 152.09 for Eclectic to Coastal transfer (average: 139.59). The asymmetry in FID scores suggests that transforming from Eclectic to Coastal style is more challenging, likely due to the simplification required to achieve the minimalist coastal aesthetic.

B. Qualitative Analysis

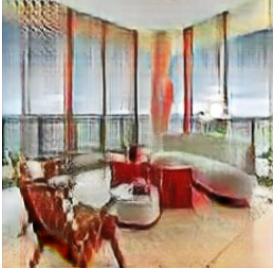
1. Coastal to Eclectic Transfer

Figures 2 and 3 show examples of transforming coastal living room styles to eclectic style.

Our model transforms coastal living rooms when converting to eclectic style by introducing significantly more



FIG. 2: Real Coastal living room (Real A)



(a) Generated Eclectic
(Fake B)



(b) Reconstructed
Coastal (Rec A)

FIG. 3: Style transfer results from Coastal to Eclectic and back

colors and vibrant patterns. We can clearly see that the generated eclectic images (Fake B) exhibit a much richer and more diverse color palette compared to the original coastal images, with the model successfully adding bold decorative elements and transforming the cool blue and white coastal palettes into the colorful schemes characteristic of eclectic design. While the overall room structure—furniture placement and architectural elements—is generally preserved during the transformation, we observe some deformations to objects and blurriness in the generated images, indicating quality limitations in the style transfer process.

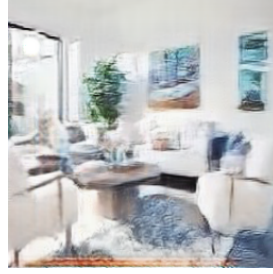
2. Eclectic to Coastal Transfer

Figures 4 and 5 show the transformation from eclectic to coastal living room style.

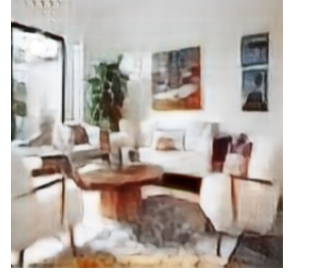
When converting from Eclectic to Coastal style, our model simplifies the color palette and reduces visual complexity. We can clearly observe that the generated coastal images (Fake A) exhibit more bluish tones, effectively transforming bold patterns and vibrant colors into the cooler, blue and white palette characteristic of coastal design. The model successfully tones down decorative elements to create the clean, airy aesthetic typical of coastal living rooms. However, similar to the opposite direction, the generated images exhibit blurriness and ob-



FIG. 4: Real Eclectic living room (Real B)



(a) Generated Coastal
(Fake A)



(b) Reconstructed
Eclectic (Rec B)

FIG. 5: Style transfer results from Eclectic to Coastal and back

ject deformations, highlighting the quality limitations of our approach.

C. Cycle Consistency Analysis

Figure 6 illustrates the full cycle of transformations for both directions.

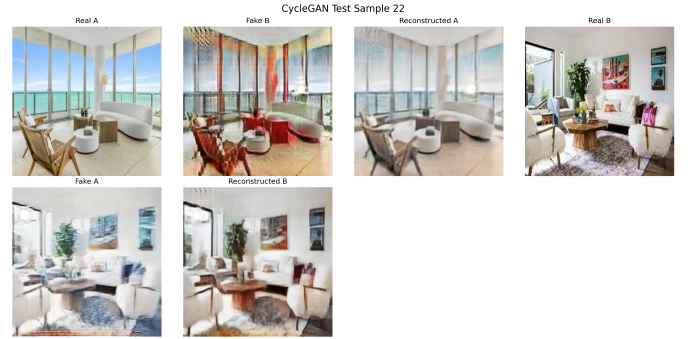


FIG. 6: Complete style transfer cycles. Coastal → Eclectic → Coastal. Eclectic → Coastal → Eclectic.

The cycle consistency results show that after a full transformation cycle, images return to a state almost similar to their originals. For the Coastal → Eclectic → Coastal cycle, the reconstructed image recovers the blue and white tones and clean aesthetic, while for the Eclectic → Coastal → Eclectic cycle, the model retrieves the

bold patterns and vibrant colors of eclectic design. However, the reconstructed images suffer from deformation and blurriness compared to the original images, indicating quality degradation through the transformation cycle despite reasonable style recovery.

D. Content Consistency Analysis

Our CycleGAN approach relies on cycle consistency loss to maintain image content while transforming style. As visible in Figures 3 and 5, the model naturally preserves room layout, furniture positions, and architectural boundaries while transforming colors, patterns, and decorative elements, though with noticeable quality degradation including blurriness and object deformations.

The cycle consistency LPIPS scores (0.3818 average) indicate that images can be reasonably reconstructed after the full transformation cycle, though these scores suggest there is significant room for improvement to achieve truly high-quality results. The asymmetric performance (better Eclectic to Coastal than Coastal to Eclectic based on FID scores) suggests that simplifying bold patterns into the blue and white coastal aesthetics may be more straightforward than the reverse transformation.

Looking at the transformation examples, we observe that furniture positions and room layouts remain largely consistent through the style transfer process, while color palettes, patterns, and decorative elements are modified to match the target aesthetic. However, both the quantitative metrics and visual results demonstrate that the current approach requires substantial improvements to deliver truly satisfactory style transfer quality. This behavior emerges naturally from the cycle consistency constraint rather than from explicit structural preservation objectives.

VI. DISCUSSION

While our CycleGAN approach produces promising results, our project has several limitations that require critical examination.

A. Limitations of Style Transfer for Interior Design

Our experiments demonstrate that the current approach can effectively transform certain stylistic elements like colors, patterns, and decorative elements, but falls short in making more substantial changes that may be desired in interior design contexts. The model primarily alters the "atmosphere" of the living rooms—adding

bold patterns when converting to eclectic style or simplifying to neutral tones for coastal style—but struggles with more comprehensive transformations.

The high FID scores (average of 192.78) indicate a significant distribution gap between generated and real images, suggesting that our transformations are not fully capturing the authentic characteristics of each design style. This limitation may come from the challenging nature of style transfer between distinctly different design aesthetics.

B. Insufficient Feature Engineering

One shortcoming of our implementation is the minimal feature engineering specific to interior design. We used a CycleGAN approach without developing domain-specific features tailored to living room style transfer. For example:

- No semantic segmentation to distinguish between different living room elements (furniture, walls, decorative items, etc.)
- Lack of style-specific transformations that could better capture design differences
- No specialized attention mechanisms to focus on style-defining elements like patterns, textures, and color schemes
- Absence of design-specific perceptual features for more meaningful loss functions

These limitations restrict the model's ability to make targeted, design-relevant transformations, instead producing more general stylistic changes.

C. Successful Style Learning and Training Dynamics

Our CycleGAN architecture showed style learning capabilities throughout training. The validation generator loss showed stable decreasing trends after the final warm restart, indicating that the generator was effectively learning to perform style transformations. The model achieved clear stylistic changes between coastal and eclectic aesthetics, transforming color palettes from cool blue and white tones to vibrant, diverse colors and vice versa. The asymmetric learning rate scheduling proved effective in preventing discriminator dominance while allowing the generator to continue improving.

The training dynamics suggest that our architecture has the potential to achieve better results with longer training periods, as evidenced by the sustained improvement in the final epochs. This suggests that with

longer training periods, hyperparameter tuning, a larger dataset, and potentially other regularization techniques, we are on the right track to achieve good results with our approach.

D. Limitations and Areas for Improvement

While our approach successfully achieves style transfer, several limitations need to be addressed. The primary challenge is image quality degradation, including blurriness and object deformations that compromise visual fidelity. Additionally, the CycleGAN’s tendency to preserve spatial content through cycle consistency, while beneficial for maintaining room layouts, limits design transformations such as:

- Replacing coastal-style furniture with bold eclectic pieces
- Generating new decorative elements not present in the original image
- Understanding the semantic meaning of different room components for targeted transformations

However, these limitations are primarily related to image quality and architectural constraints rather than fundamental failures in style learning capability.

E. Alternative Approaches

The limitations identified suggest that newer generative models, particularly diffusion models, might be better suited for interior design style transfer. Recent text-to-image and image-to-image diffusion models such as Stable Diffusion have shown superior capabilities in:

- Understanding semantic content of scenes
- Generating new objects and elements while maintaining context
- Producing high-fidelity transformations guided by text prompts
- Preserving important elements while allowing for targeted modifications

A diffusion-based approach would potentially allow for both preservation of room structure and transformation of specific design elements through controlled generation. This could enable more meaningful style transfers that actually replace bold eclectic decorative elements with clean coastal ones, or vice versa, rather than merely adjusting their appearance.

Furthermore, we could also address the fundamental limitation of our current system, which applies the same transformation logic to all room elements regardless of their design significance.

VII. CONCLUSION AND FUTURE WORK

In this paper, we presented a deep learning approach for living room interior design style transfer using a CycleGAN architecture. Our system transforms living room images between coastal and eclectic styles, changing colors, patterns, and decorative elements through adversarial training and cycle consistency loss. The quantitative results, while showing room for improvement, demonstrate that our architecture achieves meaningful style transfer. The validation generator loss showed stable decreasing trends after the final warm restart, suggesting that our architecture may work effectively through longer training periods, hyperparameter tuning, and other optimization techniques. Our visual results confirm clear stylistic transformations from cool, blue and white coastal aesthetics to bold, vibrant eclectic designs and vice versa.

The main limitations observed were image quality issues including blurriness and deformations. Our generator learned to perform style transformations, as evidenced by the clear color palette changes and stylistic modifications. Future work should focus on addressing the image quality degradation while leveraging the promising training dynamics we observed through longer training periods, hyperparameter tuning, and larger datasets. In conclusion, while our system demonstrates the feasibility of CycleGAN-based room style transfer, newer generative techniques such as diffusion models should likely outperform this method by addressing style learning, image quality, and the generation of new furniture and decorative elements that are not present in the original images.

Appendix A: Training Visualizations

This appendix contains the complete set of training visualization charts generated during our model training process. These visualizations were captured using Weights & Biases (wandb) monitoring tool over the course of 500 epochs.

1. Overall Loss Charts

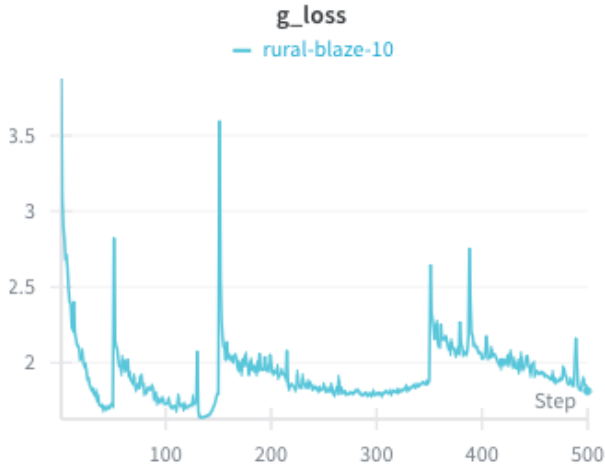


FIG. 7: Generator loss (g_loss)

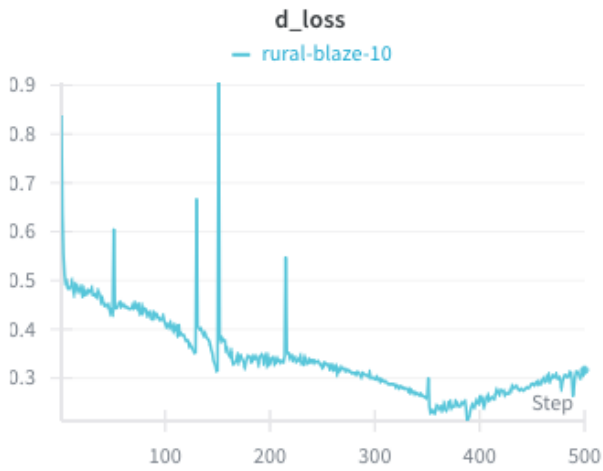


FIG. 8: Discriminator loss (d_loss)

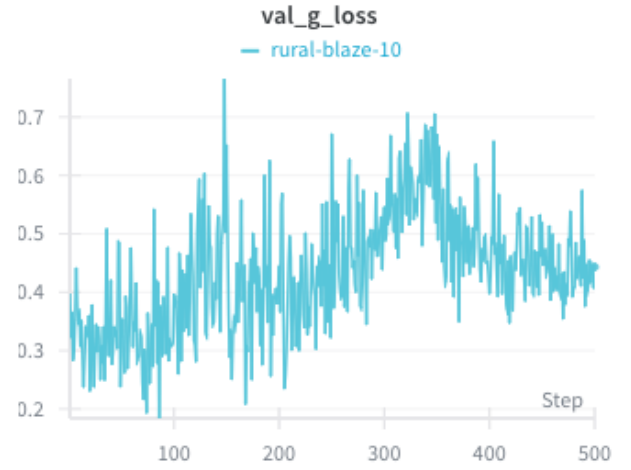


FIG. 9: Validation generator loss (val_g_loss)

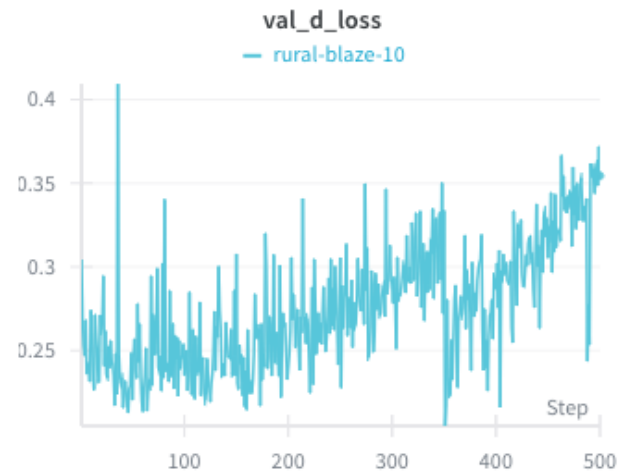


FIG. 10: Validation discriminator loss (val_d_loss)

2. Learning Rate Schedule

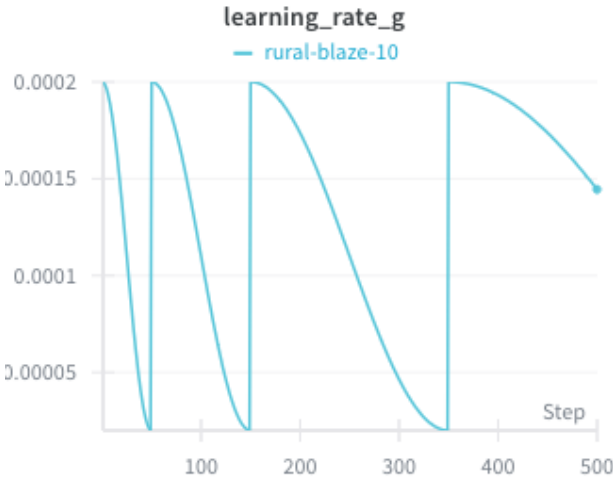


FIG. 11: Generator learning rate schedule (lr_g)

3. Domain-Specific Adversarial Losses

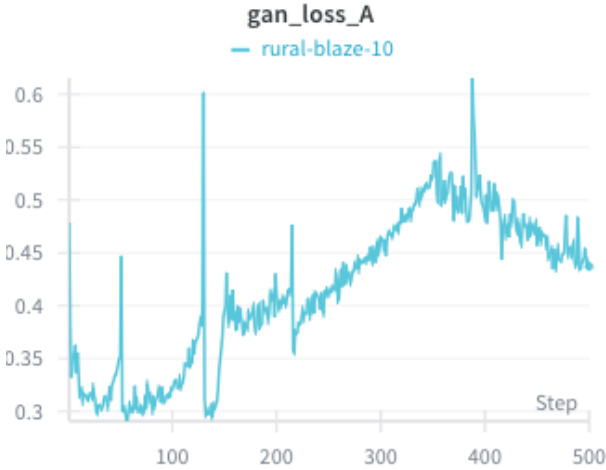


FIG. 13: GAN adversarial loss for domain A (gan_loss_A)

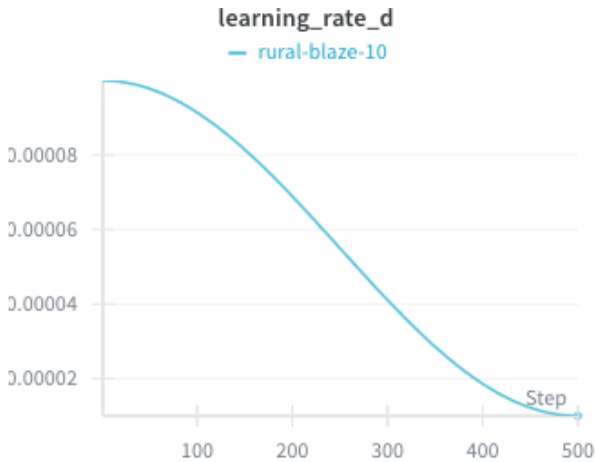


FIG. 12: Discriminator learning rate schedule (lr_d)

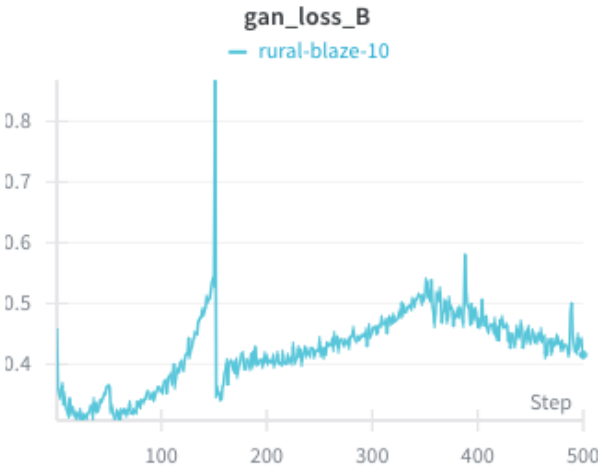


FIG. 14: GAN adversarial loss for domain B (gan_loss_B)

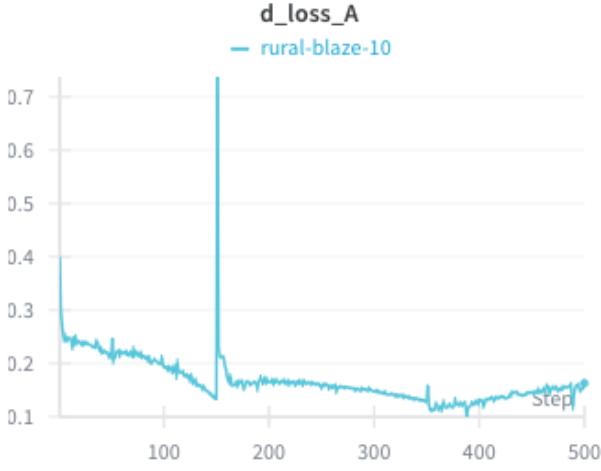


FIG. 15: Discriminator loss for domain A (d.loss_A)

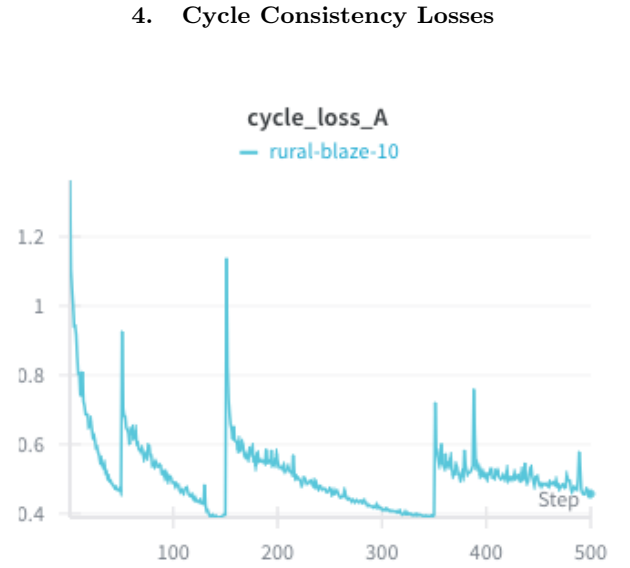


FIG. 17: Cycle consistency loss for domain A (cycle.loss_A)

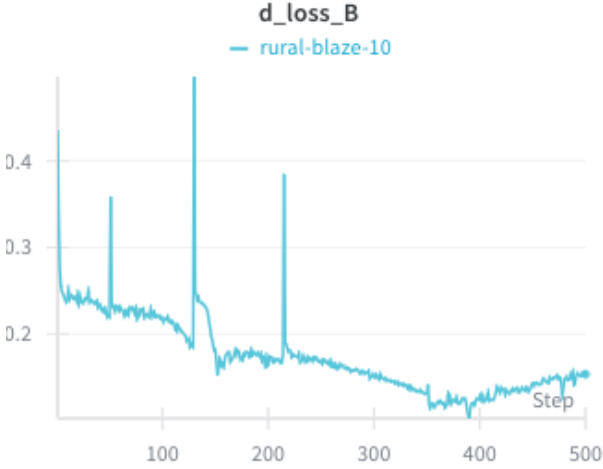


FIG. 16: Discriminator loss for domain B (d.loss_B)



FIG. 18: Cycle consistency loss for domain B (cycle.loss_B)

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